	Document No.	Section No.
	Effective Date:	Revision No.
	PROCEDURAL MANUAL	
	PREVENTIVE MAINTENANCE	

OBJECTIVE	This procedure outlines the process for scheduling, performing, documenting, and managing preventive maintenance (PM) for all heavy equipment and vehicles owned or operated by BHBI. The aim is to ensure the safe and efficient operation of equipment, minimize breakdowns, extend equipment lifespan, and reduce operating costs
SCOPE	This procedure applies to all heavy equipment used on construction sites (e.g., excavators, bulldozers, loaders, cranes, graders) and all vehicles used for transporting personnel (e.g., vans, buses, trucks) owned or leased by BHBI.

- Project Engineers: Responsible for overseeing the implementation of the PM program on their respective projects. They will coordinate with Logistics Personnel and external service providers as needed.
- Logistics Personnel: Responsible for scheduling PM activities, maintaining PM records, managing parts and supplies inventory, and coordinating with external service providers for specialized maintenance tasks. They will also be responsible for conducting basic maintenance checks and visual inspections.
- Equipment Operators: Responsible for conducting daily pre-start checks, reporting defects, and cooperating with Project Engineers and Logistics Personnel during PM activities.
- External Service Providers: Responsible for performing specialized maintenance and repairs as contracted by BHBI.

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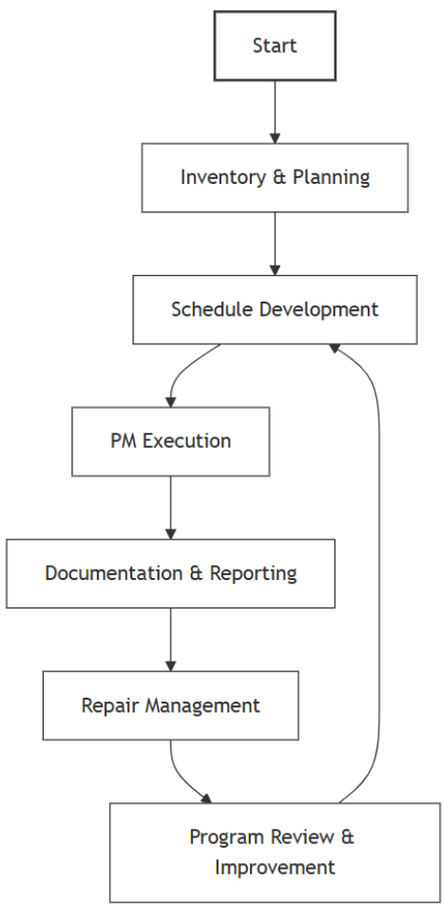

1. Procedure

1.1. Equipment Inventory and Information:

- **Maintain an up-to-date inventory of all heavy equipment and vehicles, including:**
  - **Make, model, and serial number.**
  - **Year of manufacture.**
  - **Date of purchase or lease.**
  - **Operating hours (for equipment) or mileage (for vehicles).**
  - **Assigned location or project.**
  - **Deliverables: Equipment and Vehicle Inventory**
- **Gather and organize equipment manuals (operator's manuals, service manuals) and maintenance schedules recommended by the manufacturers.**

- Deliverables: Equipment and Vehicle Manuals Library

1.2. Preventive Maintenance (PM) Schedule Development:

- **Manufacturer Recommendations: Develop PM schedules based on the manufacturer's recommendations for each piece of equipment and vehicle.**
  - **Deliverables: Manufacturer's Recommended Maintenance Schedules**
- **Usage-Based Scheduling: Schedule PM tasks based on operating hours (for equipment) or mileage (for vehicles), as well as calendar time (e.g., monthly, quarterly, annually).**
- **PM Task Lists: Create detailed PM task lists for each type of equipment and vehicle, specifying the tasks to be performed at each service interval. Clearly differentiate between tasks that can be performed by Logistics Personnel (basic checks, visual inspections) and those that require external service providers.**
  - **Deliverables: PM Task Lists**
- **PM Schedule: Develop a master PM schedule that shows all scheduled PM activities for all equipment and vehicles. Logistics Personnel will maintain and update this schedule.**
  - **Deliverables: Master PM Schedule**

1.3. PM Task Performance:

- **Work Order Generation: The assigned Logistics Personnel generates work orders or checklists for scheduled PM tasks.**
  - **Deliverables: PM Work Orders or Checklists**
- **Pre-Start Checks: Equipment operators conduct daily pre-start checks before operating equipment or vehicles, following a checklist. They report any identified issues to the Project Engineer and Logistics Personnel.**
  - **Deliverables: Daily Pre-Start Checklists**
- **PM Task Execution:**
  - **Logistics Personnel: Perform basic maintenance checks and visual inspections as outlined in the PM task lists (e.g., checking fluid levels, tire pressure, lights, basic lubrication).**
  - **External Service Providers: Engage qualified external service providers to perform specialized maintenance tasks that are beyond the capabilities of in-house personnel, as identified in the PM task lists.**
  - **Deliverables: Completed PM Checklists, Service Reports from External Providers**
- **Project Engineer Oversight: Project Engineers oversee the PM activities, ensuring that they are carried out according to schedule and that any identified issues are addressed promptly.**

1.4. Documentation and Record Keeping:

- **PM Records: Logistics Personnel maintain detailed records of all PM activities, including:**
  - **Date of service**
  - **Operating hours/mileage**
  - **Tasks performed**
  - **Parts replaced (if any)**
  - **Name of personnel or external provider who performed the work.**
  - **Any problems found and corrective actions taken.**
  - **Deliverables: PM Records, Equipment/Vehicle Service History**
- **Defect Reporting: Equipment operators report any defects or malfunctions identified during pre-start checks or operations to the Project Engineer and Logistics Personnel. Logistics Personnel maintain a log of these reports.**
  - **Deliverables: Defect Reports Log**

1.5. Repair Management (Primarily Outsourced):

- **Defect Prioritization: The Project Engineer, in consultation with Logistics Personnel, prioritizes reported defects based on severity and impact on safety and operations.**
- **Repair Scheduling: Logistics Personnel, under the guidance of the Project Engineer, schedule repairs with external service providers.**
- **Repair Oversight: Project Engineers oversee the repairs performed by external service providers to ensure quality and timely completion.**
- **Repair Records: Logistics Personnel maintain records of all repairs performed, including parts used, labor costs, and downtime, obtaining this information from external service providers.**
  - **Deliverables: Repair Records**

1.6. Parts and Supplies Management:

- **Inventory Control: Logistics Personnel maintain an inventory of commonly used PM parts and supplies (e.g., filters, lubricants, belts, hoses) if deemed cost-effective and practical.**
  - **Deliverables: PM Parts Inventory**
- **Ordering: Logistics Personnel order parts and supplies as needed, coordinating with Project Engineers and external service providers.**
- **Storage: Store parts and supplies in a clean, organized, and secure manner.**

1.7. Training:

- **Operator Training: Provide training to equipment operators on proper operation, daily pre-start checks, and defect reporting.**
- **Logistics Personnel Training: Provide training to Logistics Personnel on basic maintenance procedures, record keeping, and coordination with external service providers.**

1.8. Program Review and Improvement:

- **Regular Review: The Project Engineers and Logistics Personnel regularly review the PM program's effectiveness, analyze maintenance records, and identify areas for improvement.**
- **Performance Monitoring: Track key performance indicators (KPIs) such as PM compliance rate, equipment downtime due to breakdowns, and maintenance costs.**
  - **Deliverables: PM Performance Reports**
- **Updates: Update PM schedules, task lists, and procedures as needed based on manufacturer recommendations, operating experience, and performance data.**

Important Considerations:

- **Outsourcing:** This revised procedure heavily relies on outsourcing specialized maintenance and repairs. You'll need to have a robust process for selecting, contracting, and managing external service providers (covered in your *Procurement* and *Subcontractor Management* procedures).
- **Training:** Ensure that your Logistics Personnel receive adequate training to perform their assigned PM tasks safely and effectively.
- **Coordination:** Clear communication and coordination between Project Engineers, Logistics Personnel, Equipment Operators, and external service providers are crucial for the success of this program.
- **Documentation:** Maintain meticulous records of all PM and repair activities. This is essential for tracking equipment history, identifying recurring problems, and demonstrating due diligence.

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